

"satyam" → String

Operators & Conditionals (Logic) ✓

What are operators? + - OR, AND

→ operators are symbols or, keywords that perform an operation on values.

$5 + 3$
↓ ↘
operands operators
operands

→ unary operators → 1 operand

→ Binary ^ → 2 "

✓ typeof (variable name or, value) =

① Arithmetic operators → used for mathematical calculation.

+ → Addition

- → Subtraction

* → multiplication

/ → division

% → modulus

$$\begin{array}{r} 3 \overline{) 51} \\ \underline{3} \\ 21 \end{array}$$

→ %

$/$ $\rightarrow$ division
 $\%$ $\rightarrow$ modulus (Remainder) Remainder
 $**$ $\rightarrow$ Exponentiations $\rightarrow 2^{**}3 \rightarrow 2 \times 2 \times 2$
 $\rightarrow 5^{**}3 \rightarrow 5 \times 5 \times 5$
 $++$ $\rightarrow$ Increment
 $\hookrightarrow ++k \rightarrow k = k + 1$
 $--$ $\rightarrow$ Decrement $\rightarrow --x \rightarrow x = x - 1$

++ Assignment operators $\rightarrow$

$x = 5$ Assignment operators -

$=$

let $x = 5$

$+=$ $\rightarrow x += 2 \rightarrow x = x + 2 \rightarrow x = 7$
 $-=$ $\rightarrow x -= 2 \rightarrow x = x - 2 \rightarrow x = 3$
 $x =$ $\rightarrow x *= 2 \rightarrow x = x * 2 \rightarrow 10$

$/ =$

$\% =$

++ Comparison operators -

$\hookrightarrow$ Return $\rightarrow$ true or, false

Type casting works here

$==$ (equal) $\rightarrow$ value only ~~++~~

$\checkmark$ $===$ (equal) $\rightarrow$ value & type

$a = \underline{5}$
 $b = \underline{'5'}$

$\left[a == b \right] \rightarrow$ Type cast
 $\left[5 == 5 \right] \rightarrow$ True

$\neq$ Not equal $\rightarrow \checkmark$

$!==$ Strict not equal

$>$ Greater

$<$ less than

$>=$ Greater than equal to

$<=$ less _____

Logical operators -

$\&\&$ $\rightarrow$ Logical AND $\rightarrow$ All true $\rightarrow$ true

$||$ $\rightarrow$ Logical OR $\rightarrow$ Any true $\rightarrow$ true

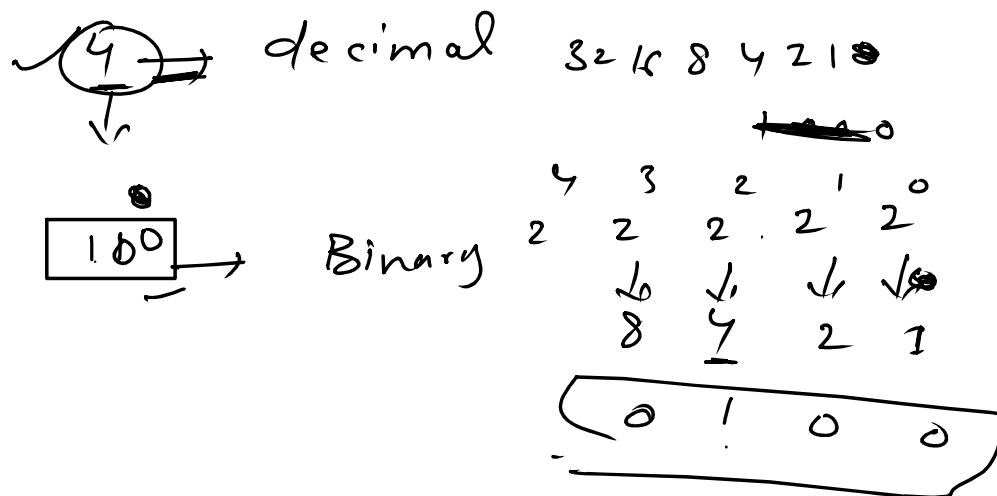
$!$ $\rightarrow$ Logical not $\rightarrow$ inverts ~~its~~ ~~to~~

True $\rightarrow$ false

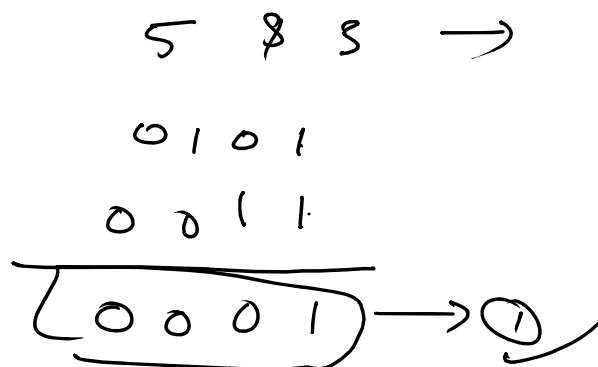
False $\rightarrow$ True

Bitwise operator $\checkmark$

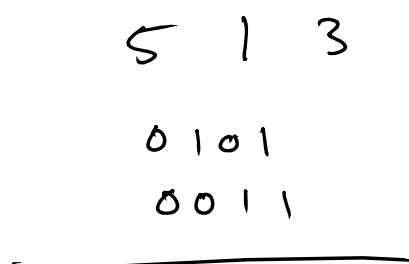
→ Bitwise operators treat their operands as a sequence of bits (0, 1) rather than decimal. Binary



→ 8 - AND → Return 1 if both 1



→ 1 → OR → either is 1 output 1



$$\underline{0111} \rightarrow (7) \checkmark$$

$\sim$ $\rightarrow$ Bitwise not

$$5 \rightarrow (\sim 5)$$

$$\rightarrow \underline{-(5+1)}$$

$$\boxed{-(5+1)}$$

$$\boxed{-(6)}$$

$$\begin{array}{r} 1111 \\ \underline{0101} \\ 1010 \end{array} \rightarrow (-6)$$

$\Rightarrow$ Left shift $\ll$

$\rightarrow$ shifts bits to the left

$\rightarrow$ multiples the no. by 2

$$a = 5$$

$$\checkmark 0101$$

$$\boxed{01010} \rightarrow (10)$$

$$5 \times 2 = 10$$

Right shift $\gg$



Right shift
divide by
half

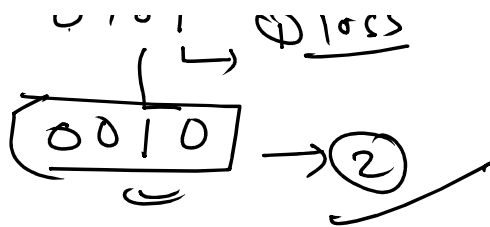
$\rightarrow$ every time you shift right by,

you can the num
in half and
drop remaineb

$$\begin{array}{r} 2.5 \\ \downarrow \\ \text{output} \\ (2) \end{array}$$

$$\underline{5 \gg 1}$$

$$\begin{array}{r} 0101 \\ \underline{} \\ 0101 \end{array} \rightarrow (2.5)$$



Conditionals / Decision making

→ Conditionals allow JS to make decision

① if ② if else ③ else if ④ switch

① if Syntax - $\left[\begin{array}{l} \text{if (} \underbrace{\text{condition}}^{\text{True}} \{ \\ \quad \text{// code} \\ \} \end{array} \right]$

② if else → $\text{if (} \underbrace{\text{condition}}^{\text{True}} \{$
 $\quad \text{// code}$
 $\} \text{ else } \{$
 $\quad \text{// code}$
 $\}$

⑪ else if

- if(condition) {
 // code ✓

3 else if(condition) {

3 // code ✓

else if (condition) {

 // code ✓

3

else {

 // code

3

if else
↓
if else
↓
if else

→ Switch → Switch is another way to handle multiple specific possible values.

Syntax -

switch () {

case 1 " " :

 " " .

```

    case 2:
    {
        //
        //
    }
    default:
    {
        //
    }
}

```

Ternary operator -

→ The ternary operator is a short way of writing a simple if...else conditions

Syntax -
 [condition ? value if true : value if false]